

Audio-R1 Training Pipeline

Raw Audio $\mathbf{a}$



Predefined SSL Tasks

Transformed Audio $\mathbf{a}'$
with Derived Answers y^*

Audio Language Model π_θ

$\bar{y}_1$



$\bar{y}_2$



$\bar{y}_3$

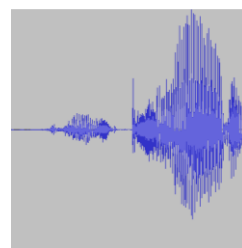


Answer Verification

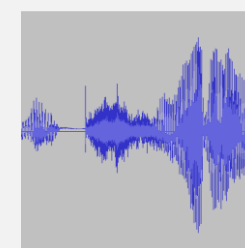
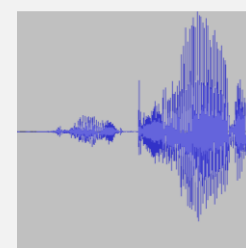
Reward-Guided Policy Update

$$r_i = \mathcal{V}(y_i, y^*)$$

Identity Detection



Query

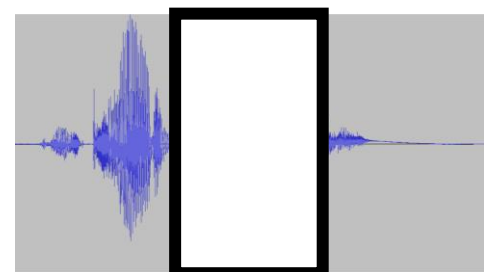


Candidates

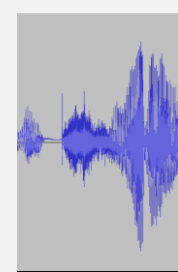
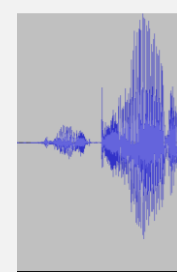
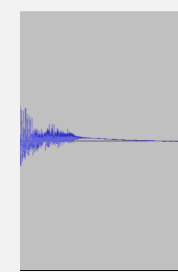
Q. Which matches?

- A. First
- B. Second
- C. Both
- D. Neither

Region Inpainting



Masked Audio

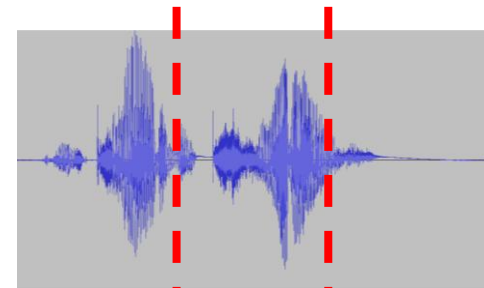


Candidates

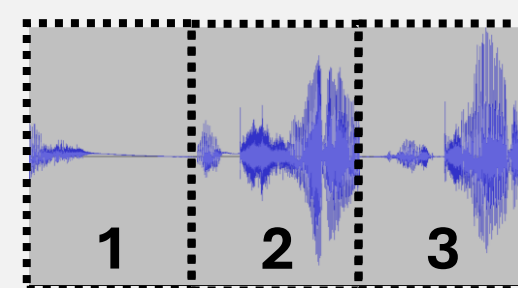
Q. Which fits?

- A. First
- B. Second
- C. Third
- D. Neither

Temporal Ordering



Original Audio



Shuffled Audio

Q. Which order?

- A. 1 – 2 – 3
- B. 2 – 1 – 3
- C. 2 – 3 – 1
- D. 3 – 2 – 1